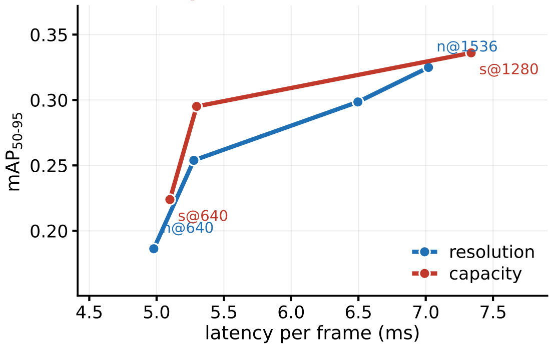
Where should a fixed, measured inference budget go: resolution or capacity?

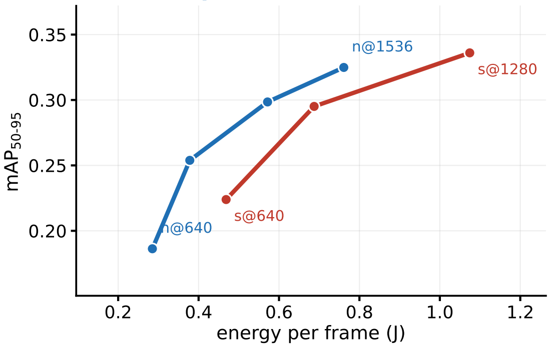
Real-time UAV detection, with cost measured on the device rather than inferred from parameters or FLOPs

Main result — VisDrone

capacity leads by 0.0248
compared over 5.19–6.95 ms



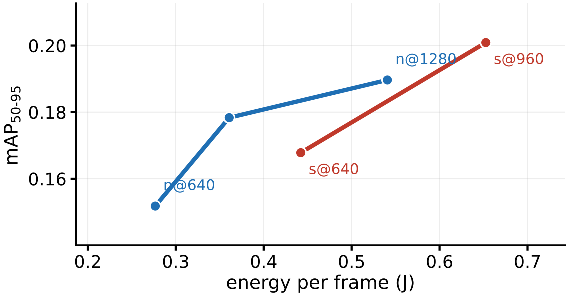
resolution leads by 0.0331
compared over 0.48–0.76 J



Robustness: with the device's idle draw removed, resolution leads by +0.0457 over 0.26–0.46 J

Second dataset — UAVDT

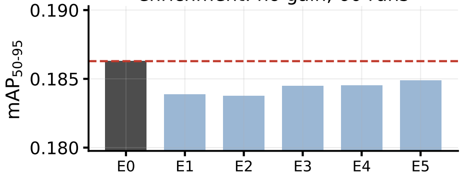
same directions, wider spread



Both directions replicate.
latency -0.0245   energy +0.0112
Neither reaches corrected significance:
seed variance is about seven times higher
and the arms barely overlap.

Three pre-registered eliminations came first

enrichment: no gain, 60 runs



1 Retrieval-based data enrichment
no gain at any budget tested

2 Size-conditioned confidence correction
little to recover: the low confidence is accurate

3 Our own first hypothesis
rejected once cost was measured, not assumed

Takeaway

Resolution alone lifts $mAP_{50\text{-}95}$
from 0.1863 to 0.3249 between 640
and 1536 px, 74% relative, concentrated
on small objects.

Latency-bound → spend on capacity
Energy-bound → spend on resolution

Report both axes, measured on the device.